

EOQ Sensitivity Analysis

Lecture item: $A=3.2$, $D = 2,400$, $v = 0.4$, $r=0.24$

Inventory and Production Management

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Percentage cost penalty

$$Q' = (1 + \alpha)EOQ$$

$$PCP = \frac{TRC(Q') - TRC(EOQ)}{TRC(EOQ)} \times 100 = 50 \frac{\alpha^2}{1 + \alpha}$$

The PCP depends only on alpha: it is the same for every item. The curve is flat near alpha = 0 and asymmetric: ordering too little costs more than ordering too much by the same percentage.

A parameter estimated as f times its true value gives $1 + \alpha = \sqrt{f}$ (for D or A) or $1/\sqrt{f}$ (for v or r).

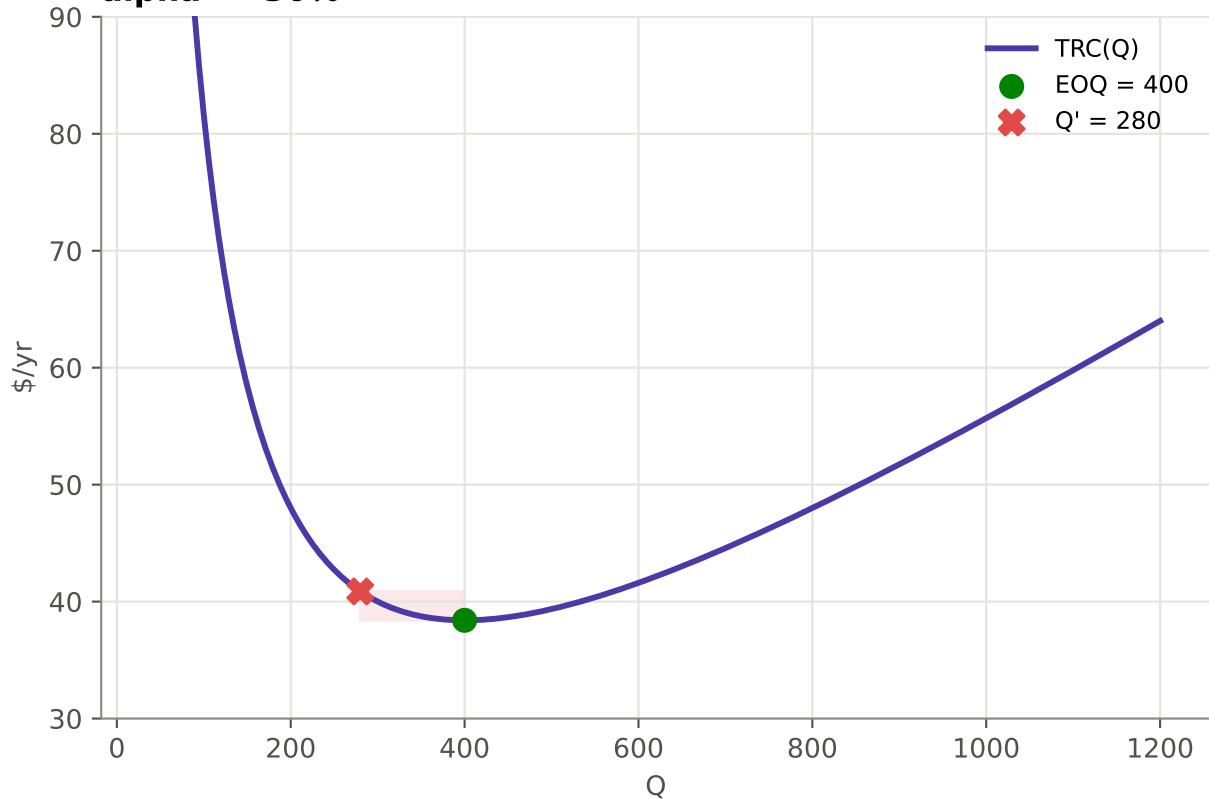
PCP for selected deviations

alpha	Q'/EOQ	PCP (%)
-0.5	0.5	25
-0.3	0.7	6.429
-0.2	0.8	2.5
-0.1	0.9	0.556
0	1	0
0.1	1.1	0.455
0.2	1.2	1.667
0.3	1.3	3.462
0.5	1.5	8.333
1	2	25

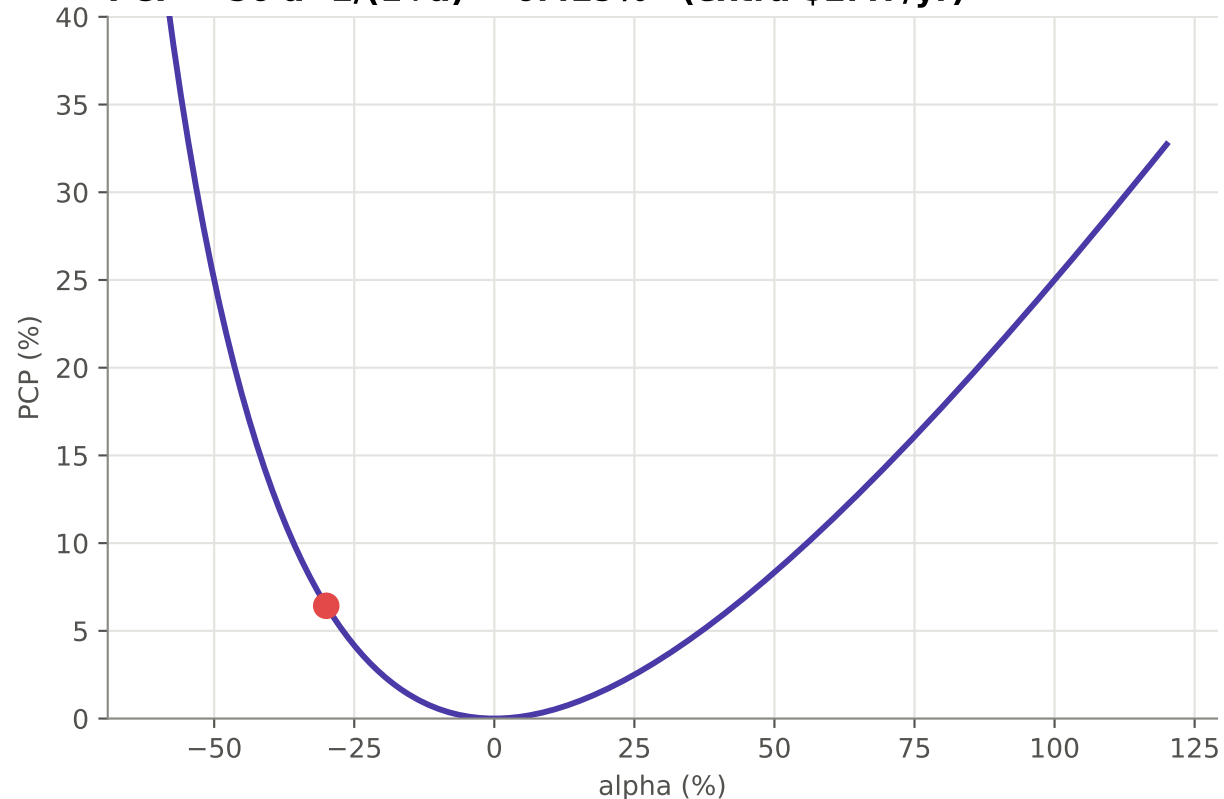
Practical policies

scenario	Q_used	alpha	PCP (%)	Extra cost (\$/yr)
Order exactly the EOQ	400	0	0	0
Supplier ships in cases of 250	250	-0.375	11.25	4.32
Supplier ships in cases of 500	500	0.25	2.5	0.96
Buyer orders one month of demand	200	-0.5	25	9.6
Buyer orders one quarter of demand	600	0.5	8.333	3.2
Warehouse orders half a year at a time	1,200	2	66.67	25.6
Buyer orders weekly	46	-0.885	340.5	130.8

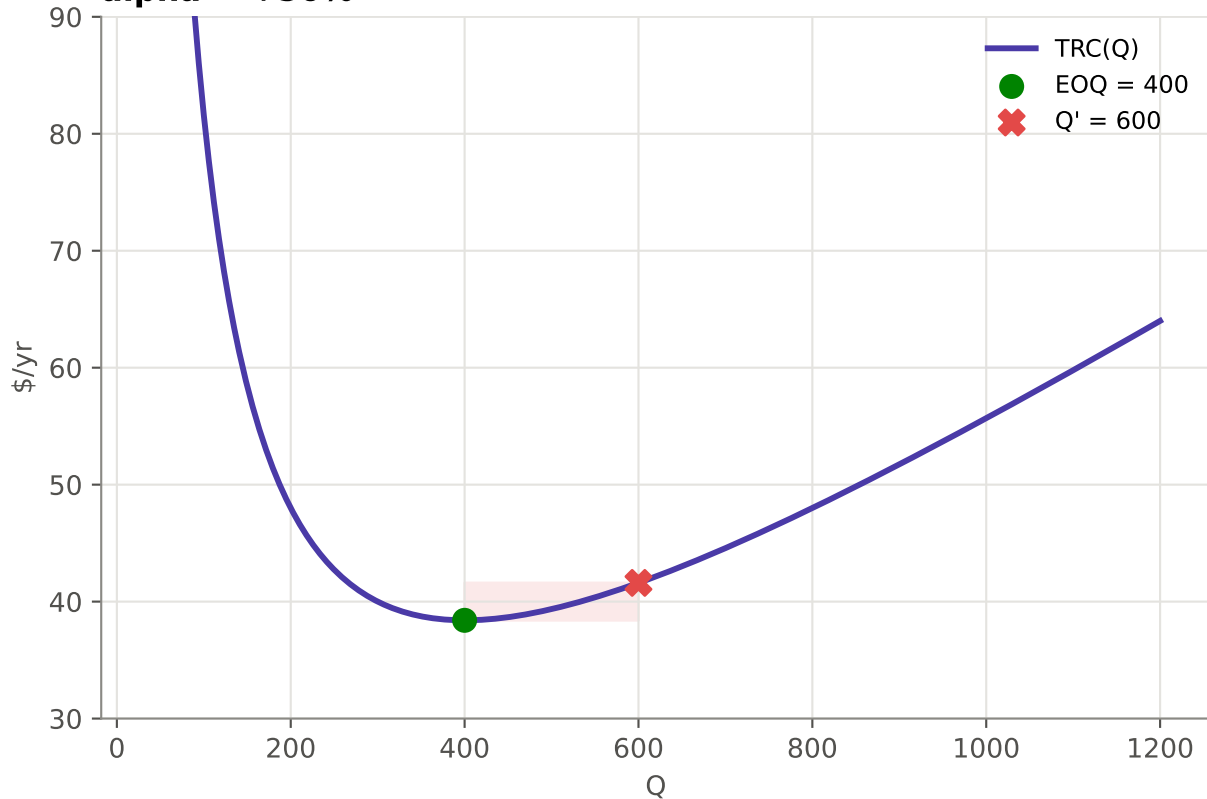
alpha = -30%



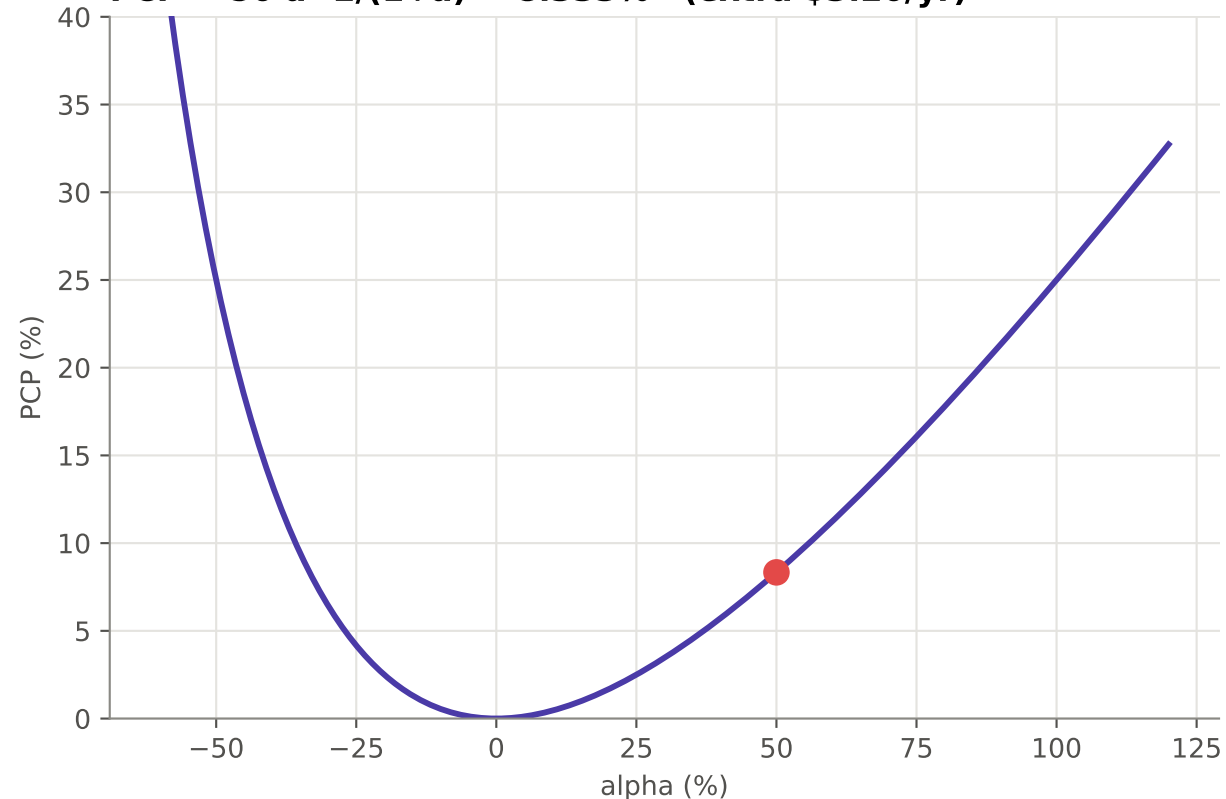
PCP = 50 $a^2/(1+a)$ = 6.429% (extra \$2.47/yr)



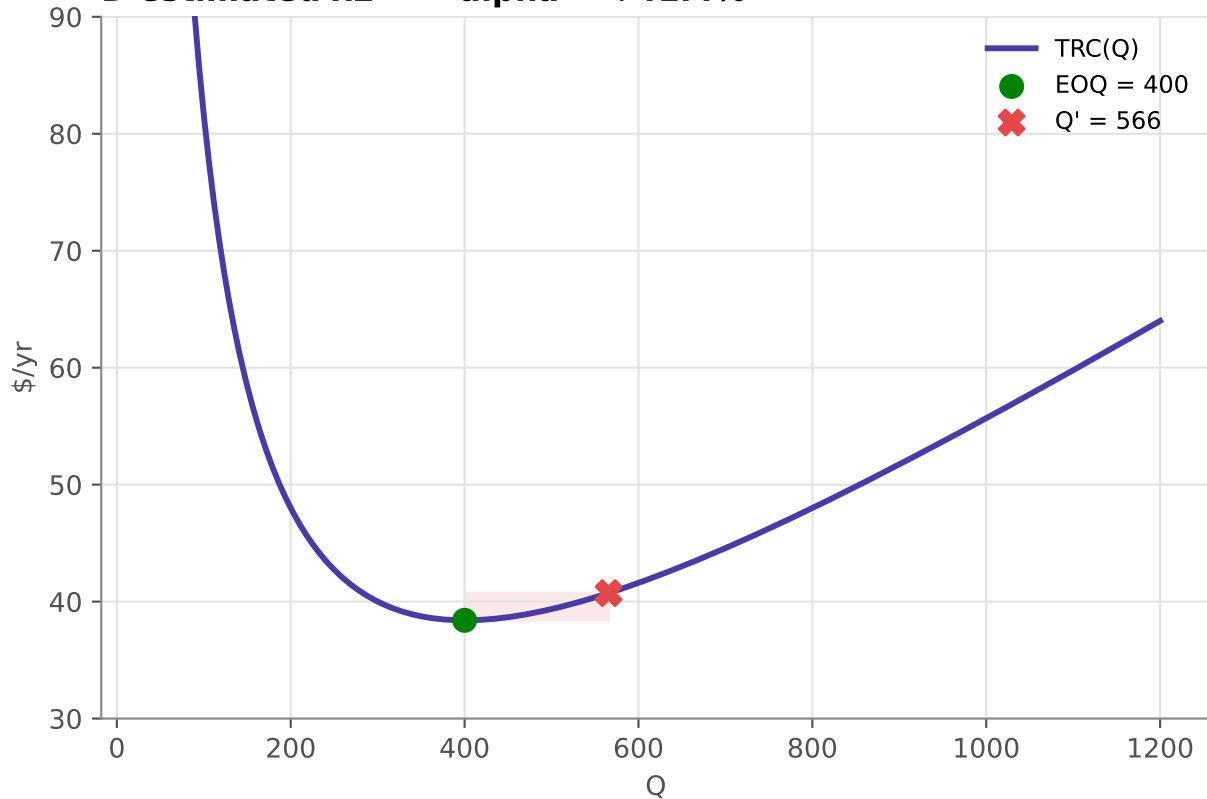
alpha = +50%



PCP = $50 \frac{a^2}{(1+a)} = 8.333\%$ (extra \$3.20/yr)



D estimated x2 -> alpha = +41.4%



PCP = 50 $a^2/(1+a)$ = 6.066% (extra \$2.33/yr)

